

Vikram Anantha

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Education

University of Michigan

Aug. 2024 – May 2028 (expected)

B.S. Computer Science & Electrical Engineering

3.93/4.00 GPA

- **Relevant coursework:** Logic Design, SLAM, Machine Learning, Computer Vision, Data Structures & Algorithms, Operating Systems, Digital Circuits, Computer Organization, Linear Algebra, Discrete Math, Differential Equations
- **Clubs:** Head Photo Editor of the Michiganensian Yearbook, Active member of Tau Beta Pi Engineering Honors Society
- **Honors:** 1/6 recipients of 1931e Merit Scholarship for Engineering, 3-term James B. Angell Scholar, Tau Beta Pi Scholarship

Lexington High School

Sept. 2020 – Jun. 2024

Lexington, MA

4.00/4.00 GPA

- 36/36 ACT score • 5/5 on AP Computer Science A, AP Calculus BC, AP Physics C, AP Chemistry, AP Statistics, AP Biology

Industry Experience

Meta

May 2026 – Jul. 2026

Software Eng. Intern

Menlo Park, CA

- Architected first Server-Side Rendering pipeline for React Native at Meta, cutting client load
- Reduced client bundle size 30% and presented findings org-wide to drive RSC adoption across apps

Capoom

Jan. 2025 – Dec. 2025

Software Eng. Intern, via Perot Jain Tech Lab at U of M's Center for Entrepreneurship

Ann Arbor, MI

- Benchmarked Gaussian Grouping, LangSplat, SAM against real Mcity/Ford sensor data for AV training
- Delivered segmentation pipeline to production while leading a team of 3 student engineers

Stellantis (F.K.A. Fiat-Chrysler Automotive)

May 2025 – Aug. 2025

ADAS Functional Integration Eng. Intern

Auburn Hills, MI

- Engineered CAPL script to analyze CAN-bus signals, surfacing a 10% fault rate for ADAS teams
- Built MATLAB Simulink model to validate Level 2 autonomy CAN-bus behavior for ADAS integration

Constant Therapy Health

Jun. 2024 – Aug. 2024

Software Eng. Intern

Newton, MA

- Fine-tuned Wav2Vec ASR on noisy aphasia phonemes, achieving 4x lower false negative rate than Google's
- Built task visualizer and LLM grammar detector to support stroke patient speech therapy

KTBYTE Computer Science Academy

Apr. 2018 – May 2024

Lead instructor, Web+Curriculum Developer

Lexington, MA

- Lead instructor & curriculum developer of Python, Java, Arduino classes for 100+ students, ages 8-18
- Robotics lead instructor, mentored students to create inventions w/ Arduino+Raspberry Pi; 15+ student awards nationally

Selected Projects

Eous AR-Car

Nov. 2025

- Built AR-glasses assistant enabling gesture-based robot control and live video/audio streaming, 1st place at Hackathon
- Engineered thread-safe Unity HTTP server and Python backend for video encoding, transcription

HELM Learning

Apr. 2020 - Present

- Created Peer-to-peer free online learning platform for K-12 students; published, arXiv:2209.03489
- Scaled Flask/AWS/MySQL backend for 90+ classes, 4.2k+ students worldwide, course recom. algo w/ 93% accuracy

Quantum-Enhanced ML Algorithm to Detect Alzheimer's

Oct. 2021 - May 2022

- Co-built MLP/CNN ensemble using Quantum Preprocessing to diagnose Alzheimer's with over 85% accuracy
- Validated by neuroscience experts at MIT and Harvard; won MSEF Regionals and States 2022

My Automated Communication Companion

Feb. 2019 - Jun 2021

- Built smart-hat using object detection and speech recognition to teach non-verbal kids to speak
- Tested with a non-verbal autistic child, who successfully repeated a word using the device

Research Experience

University of Michigan, ROAHM Robotics Lab	Aug. 2024 – Present
- Built real-time 3D scene pipeline (Gaussian Splat, SAM-2, Grounding DINO) on ROS2 for Fetch arm - Building feed-forward ViT-to-SDF network for sub-second, collision-aware 3D scene queries for VLA robots, targeting ICRA	
Columbia University, Center for Smart Streetscapes	May 2023 – Feb. 2024
- Built camera-based localization + MQTT pedestrian alert system; co-authored 2024 IEEE IV paper - Optimized alert pipeline to <1.5s latency, meeting human reaction time to prevent accidents	
Massachusetts Institute of Technology, Auto ID Lab	Jun. 2023 – Sep. 2023
Built YOLOv8 + binary-image 2FA system to verify AV headlight patterns, securing V2I communication	
Massachusetts Institute of Technology, MetaConscious Group	Jul. 2022 – Aug. 2022
Created RL agent modeling primate learning under a changing environment	

Photography Experience

Michiganensian Yearbook	Sept. 2024 – Present
<i>Head Photo Editor</i>	<i>University of Michigan, Ann Arbor, MI</i>
- Manage photographic workflow, organize assignments, deadlines, credentials - Lead meetings for development and critiques, proof and re-edit for aesthetic coherence	
Lexington Observer	Jun. 2023 – Present
<i>Staff Photographer</i>	<i>Lexington, MA</i>
Official Photographer for the Town of Lexington MA's online journal; 50+ photos; My articles: lexobserver.org/author/vikram-anantha	
Town of Lexington, Lexington Public Schools	Jun. 2023 – Present
<i>Freelance Photographer</i>	<i>Lexington, MA</i>
Official Photographer for the Lexington MA's town government and public schools; 100+ photos	
Lexington High School Yearbook + newspaper <i>The Musket</i>	Sept. 2022 – Jun. 2024
<i>Head Photo Editor</i>	<i>Lexington High School, Lexington, MA</i>
- Managed photographic workflow, organized assignments, deadlines, credentials - Lead meetings for development and critiques, proofed and re-edited for aesthetic coherence	

Technical Skills

Programming Languages: Python, C++, C, Java, JavaScript, CAPL, ReactJS, Assembly, Bash, LaTeX
Frameworks & Tools: ROS/ROS2, Linux, AWS, Vector CANalyzer, MATLAB/Simulink, OpenCV, TensorFlow/PyTorch, Git/GitHub, CUDA, Docker, Gaussian Splatting, LLMs, YOLOv8, SAM-2, SLAM

Selected Honors & Awards

Immerse The Bay Hackathon, Stanford University, 1st Place Winner	Nov. 2025
Lexington Public Schools Core Values award, Reciept, awarded by LPS superintendent	Oct. 2023
Diamond Challenge Business Competition, 1st Place Winner for HELM Learning	Mar. 2023
Drexel University High School Photo Contest, Selected for Photo Exhibition	Dec. 2022 & 2023
Kurt Giessler Foundation, Prize for Achievement for HELM Learning (\$750 grant money)	Oct. 2022
Mass. Science & Engineering Fair (MSEF), Outstanding AI Project Winner	Mar. & May 2022
American Invitational Mathematics Exam (AIME), 2 Time Qualifier	Mar. 2021 & Feb. 2022
Presidential Volunteer Service Award (PVSA), Gold Award Recipient	Nov. 2021
USA Coding Olympiad (USACO), Silver Division	Dec. 2019
Invention Convention Worldwide (ICW), UTC Best in Show: Most Innovative Project	May 2019

Publications & Academic Service

CVPR Workshop for Autonomous Driving (WAD), Peer Reviewer	2026
Digital Twin for Pedestrian Safety Warning at a Single Urban Traffic Intersection	2024
V. Anantha, et al., 2024 IEEE Intelligent Vehicles Symposium (IV), pp. 2640-2645, doi: 10.1109/IV55156.2024.10588544	
Peer to Peer Learning Platform Optimized With Machine Learning	2022
V. Anantha, arXiv:2209.03489, 2022	